

# 本周进展

- UE内进展

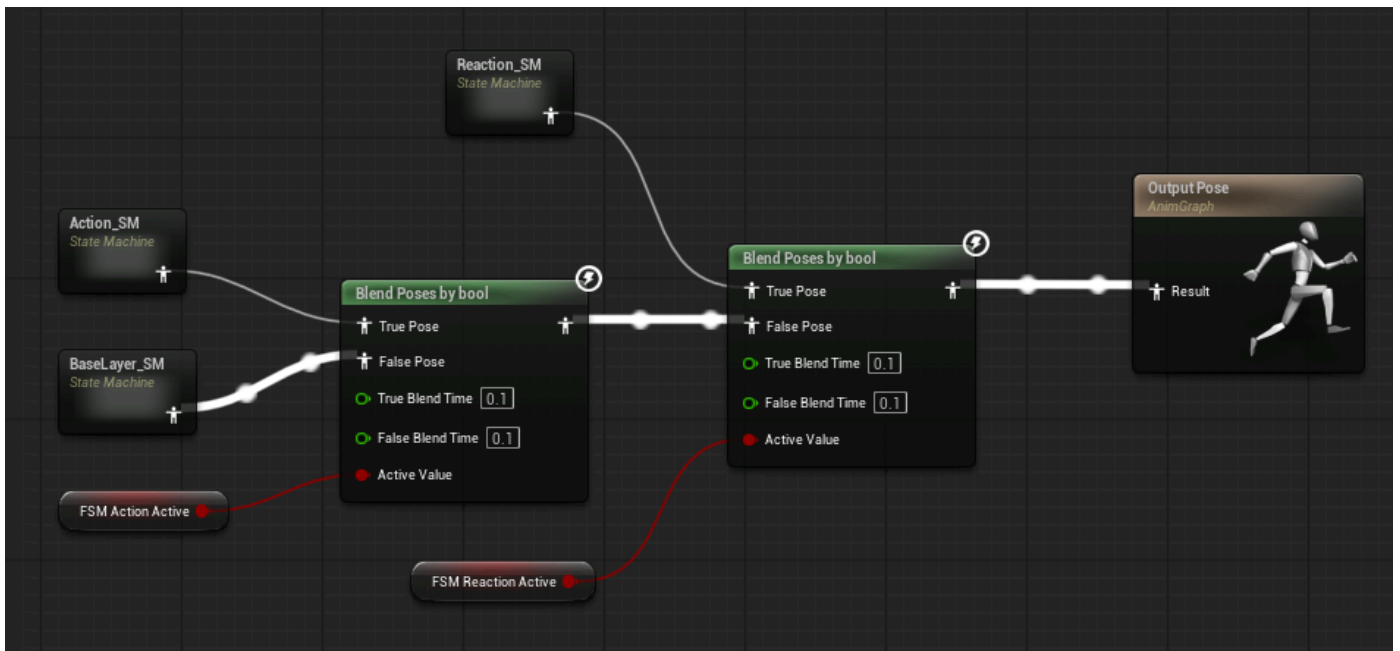
UE内置lua脚本系统，仿照只狼动画控制脚本，制作了八方向移动逻辑, 动画丝滑度还需完善

## 动画状态机设计梳理

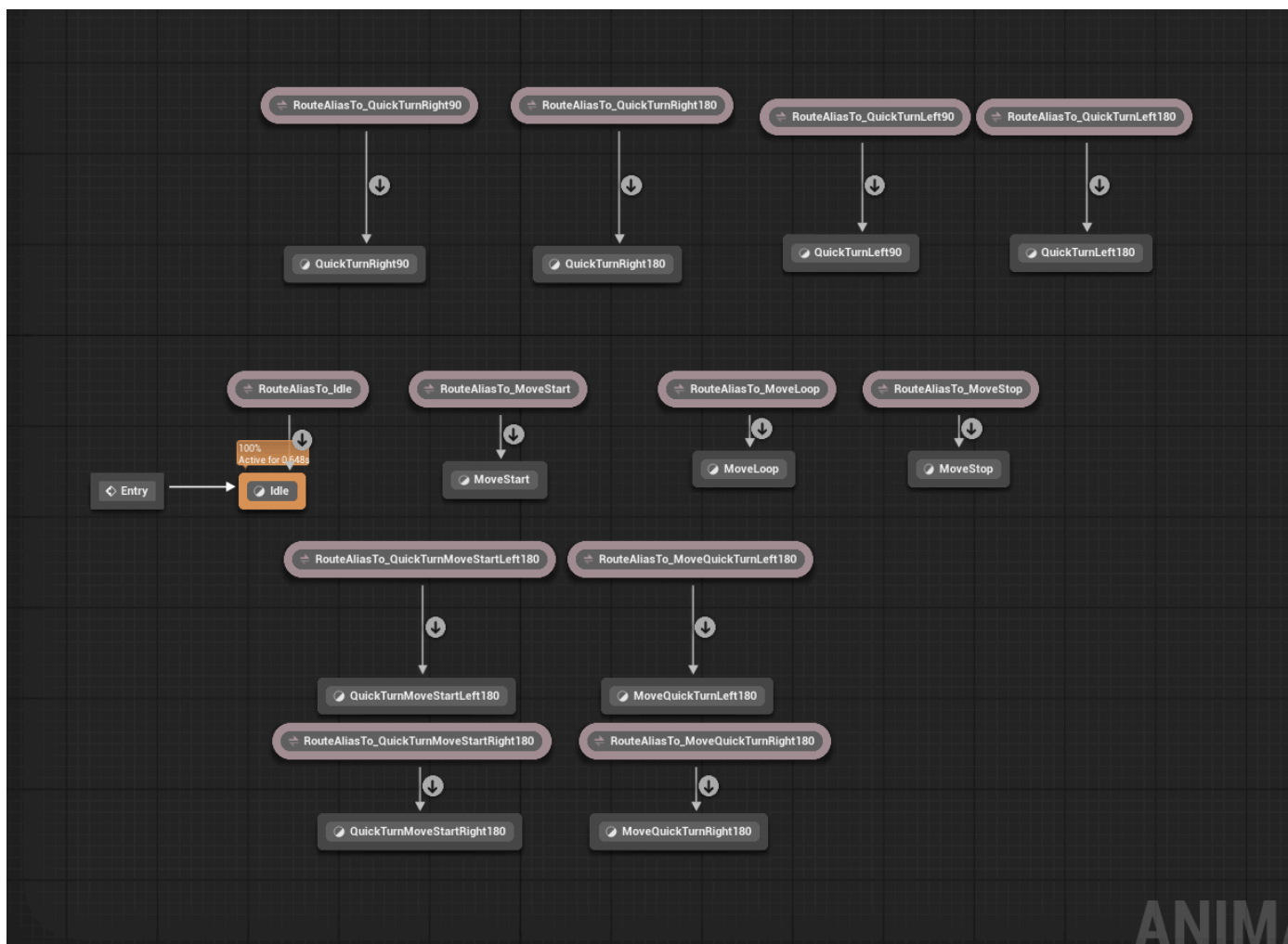
1. 动画状态机参照只狼的设置分Style和State，Layer，避免状态机太乱，可以根据不同的Style在内部进行状态筛选
  - **Style**: 常态，战斗姿态，居合，倒地等
  - **State**: Idle, Run, Walk, TurnLeft90等
  - **Layer**: 上半身（持刀，挥刀，格挡），下半身(走，跑，原地转身)，附加层(披风挂件，武器)
2. 最上层所有的状态又分为Action（主动反应）和Reaction（被动反应），reaction优先级更高，被动反应可以打断主动动作，比如说被击飞打断攻击

## UE内状态机设计

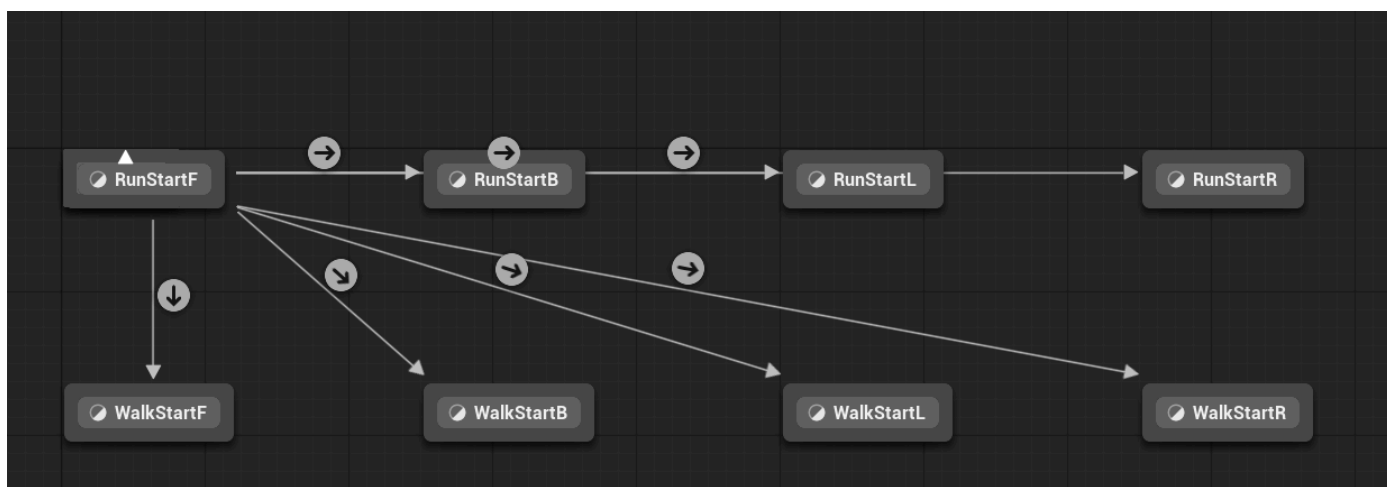
1. 设计了Reaction，Action,和Base，默认是执行Base，Reaction和Action暂时空置



2. Base层里面目前只有常态



3. 具体状态下面设计了selector选择不同朝向的动画



根据不同的朝向输入选择移动启动动画, 这些节点内部暂时还未实现上下半身分层动画

## 动画蓝图C++和lua动画变量交互设计

1. 只狼状态更新

```

function UpdateState(current_hkb_state)
    Control(current_hkb_state)
    Validate(current_hkb_state)
    FireStateEndEvent(current_hkb_state)
end

```

其中的Control将决定当前状态属于哪个抽象姿态(style)

Validate决定里面执行是Action还是Reaction还是base

## 2. 仿照只狼处理动画变量

```

function M:ReceiveTick(delta_seconds)
    local runtime = ensure_runtime(self)
    runtime.time = runtime.time + math.max(resolve_number(delta_seconds, 0.0), 0.0)

    sync_polled_input(self)
    local context = read_context(self, delta_seconds)
    validate_layer(self, Constants.LAYER_REACTION, context)

    context = read_context(self, delta_seconds)
    validate_layer(self, Constants.LAYER_ACTION, context)

    context = read_context(self, delta_seconds)
    validate_layer(self, Constants.LAYER_BASE, context)

    context = read_context(self, delta_seconds)
    if face_toward_move_input(self, context) then
        context = read_context(self, delta_seconds)
    end
    apply_anim_values(self, context)
    apply_movement_input(self, context)
    flush_request_pulses(self)
    render_debug_overlay(self, context)

    if self.Overridden and self.Overridden.ReceiveTick then
        self.Overridden.ReceiveTick(self, delta_seconds)
    end
end

```

# Lua和C++之间关于动画变量的交互

C++端定义一堆变量查询，这个是根据现有资料猜测的变量

```

const TArray<FSekiroEnvIdInfo>& USEkieroEnvQueryComponent::GetEnvInfoTable()
{
    static const TArray<FSekiroEnvIdInfo> Infos = []() ->TArray<FSekiroEnvIdInfo>
    {
        TArray<FSekiroEnvIdInfo> Result;
        Result.Reserve( Number: 82);
        Result.Add( Item: MakeEnvInfo( Id: 105, Key: TEXT("env(105, slot)"), Name: TEXT("PendingEventId"), Meaning: TEXT("Pending EZ/event id."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 113, Key: TEXT("env(113)"), Name: TEXT("ItemUseFixedRequest"), Meaning: TEXT("Special fixed-item branch before item use request."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 115, Key: TEXT("env(115)"), Name: TEXT("ItemUseRequestInvalid"), Meaning: TEXT("Item request is blocked or invalid."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 200, Key: TEXT("env(200)"), Name: TEXT("FallRequest"), Meaning: TEXT("Fall transition request."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 201, Key: TEXT("env(201)"), Name: TEXT("LandRequest"), Meaning: TEXT("Land/contact transition request."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 202, Key: TEXT("env(202)"), Name: TEXT("DamageType"), Meaning: TEXT("Damage type enum."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 205, Key: TEXT("env(205)"), Name: TEXT("DamageAnimationGate"), Meaning: TEXT("Damage/reaction gate used with 1112/205 damage type."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 206, Key: TEXT("env(206)"), Name: TEXT("DamageReactionSuppressed"), Meaning: TEXT("Suppress death/damage/docking-timer when damage is received."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 207, Key: TEXT("env(207)"), Name: TEXT("WeaponChangeType"), Meaning: TEXT("Weapon/arm style, compared with ARM_STYLE enum."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 222, Key: TEXT("env(222)"), Name: TEXT("DamageAngle"), Meaning: TEXT("Horizontal hit direction."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 224, Key: TEXT("env(224)"), Name: TEXT("FallHeightRaw"), Meaning: TEXT("Raw fall-height value; Lua multiplies by 0.5 to get actual height."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 225, Key: TEXT("env(225, HAND_*)"), Name: TEXT("WeaponMotionCategory"), Meaning: TEXT("Weapon motion category by hand."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 231, Key: TEXT("env(231)"), Name: TEXT("ItemAnimeType"), Meaning: TEXT("Item animation type."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 233, Key: TEXT("env(233)"), Name: TEXT("ItemUseEnable"), Meaning: TEXT("Current item can be used."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 236, Key: TEXT("env(236)"), Name: TEXT("DamageLevel"), Meaning: TEXT("Damage level enum."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 237, Key: TEXT("env(237)"), Name: TEXT("GuardDamageAmount"), Meaning: TEXT("Guard damage or posture damage amount."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 248, Key: TEXT("env(248)"), Name: TEXT("LandReady"), Meaning: TEXT("Ground/Land-ready flag."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 256, Key: TEXT("env(256)"), Name: TEXT("HpAutoChargeActive"), Meaning: TEXT("HP auto-charge related state."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 273, Key: TEXT("env(273)"), Name: TEXT("ThrowAnimationId"), Meaning: TEXT("Throw animation id."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 274, Key: TEXT("env(274)"), Name: TEXT("ThrowKillRequested"), Meaning: TEXT("Throw kill request."), ArgumentFamily: TEXT("none")),
        Result.Add( Item: MakeEnvInfo( Id: 276, Key: TEXT("env(276)"), Name: TEXT("ThrowDeathRequested"), Meaning: TEXT("Throw death request."), ArgumentFamily: TEXT("none"))
    };
}

```

Lua端通过env(105)即可查询这个变量值

Lua端通过更新这些数据可以改变C++端动画蓝图变量

```

sync_anim_state_vars(self, layer)
sync_add_layer_anim_vars(self)
set_anim_float(self, "MoveSpeedLevel", move_speed_level)
set_anim_float(self, "MoveSpeedLevelReal", move_speed_level_real)
set_anim_float(self, "MoveLoopPlayRate", move_loop_play_rate)
set_anim_int(self, "MoveSpeedIndex", move_speed_index)
set_anim_int(self, "NightvisionMoveSpeedIndex", move_speed_index + 1)
set_anim_int(self, "FSM_MoveStartSelectorId", move_start_selector_id)
set_anim_float(self, "MoveDirection", direction)
set_anim_int(self, "MoveDirectionIndex", direction)
set_anim_float(self, "MoveAngle", move_angle)
set_anim_float(self, "TurnAngle", turn_angle)
set_anim_float(self, "TwistLowerRootAngle", twist_angle)
set_anim_int(self, "MoveType", move_type)
set_anim_int(self, "StanceMoveType", stance_move_type)
set_anim_bool(self, "bSprintHeld", context.sprint)

```

只狼的设计是将不同的状态切换条件按照[style][state]来存储，根据引擎内部触发不同的条件查询不同的env变量，查询出切换条件，会设置到C++这边来切换动画状态。

## 下周目标

- 简化移动动画状态机
- 还原只狼的移动事件触发逻辑，还原移动丝滑度